ABDESSAMAD AIT BRAHIM



AI Engineering Student | Machine Learning, NLP & Computer Vision

Seeking a 6-month final-year internship (PFE) in AI starting January 2027

abdessamadaitbrahim9@gmail.com **+212 6 43 14 25 99** **Morocco**

linkedin/abdessamad-aitbrahim **github/abdessamadaitbrahim** **Portfolio**

PROFILE

Final-year engineering student in **Big Data & Artificial Intelligence** at ENSA Tétouan (Morocco), **seeking a 6-month final-year internship (PFE) in AI starting January 2027**. Three AI engineering internships delivering practical solutions: an **LLM conversational agent**, **multilingual NLP** (XLM-RoBERTa, mBERT, DarijaBERT) and **deep-learning candidate matching**. Strong in experimental comparison and rigorous evaluation across machine learning, deep learning, NLP and computer vision.

PROFESSIONAL EXPERIENCE

AI Engineer Intern — LLM Conversational Agent (POC) Aug – Sep 2026

STMicroelectronics · Bouskoura, Casablanca, Morocco

- Designed an **LLM-powered conversational agent** that makes real-time manufacturing performance data accessible in natural language, extending an existing production-capacity monitoring tool.
- Built a **Python/pandas** pipeline to clean, validate and transform **10,000+ production records** into reliable performance indicators; isolated and resolved a critical data-quality issue through hypothesis-driven testing.
- Architected the system with **MariaDB**, a secured read-only **FastAPI** service and **Dify** for LLM orchestration; engineered a strict system prompt so the agent never fabricates data and flags reliability limits.
- Delivered a working demo answering real engineering questions (underperforming machines, bottlenecks, throughput improvements).

AI Engineer Intern — Labex / Botground Jun – Jul 2026

SaaS Project, Multilingual Sentiment Analysis · Casablanca, Morocco

- Built an end-to-end **NLP pipeline** classifying chatbot conversations in French, English and Moroccan Darija.
- Fine-tuned and benchmarked **XLM-RoBERTa, mBERT and DarijaBERT** under identical conditions to justify model selection, reaching **84.3%** accuracy.
- Generated and curated a synthetic multilingual dataset of **4,000+ annotated conversations** across hospitality, logistics and customer support.
- Developed a **Flask/SQLite** analytics dashboard (KPI tracking, sentiment trends, problematic-conversation detection) and deployed it on **AWS EC2/S3**.

AI Engineer Intern — SmartMatchRH Jul – Aug 2025

Cloud Marketing Hub · Tangier, Morocco (On site)

- Developed a **candidate/job matching** model using deep learning and multilingual NLP (Python, TensorFlow) for CV analysis and scoring.
- Built an end-to-end data pipeline (ingestion, cleaning, feature engineering) served through a **Flask REST API**, with an HR chatbot and **3 Power BI dashboards** for KPI tracking.
- Reported results: **+35%** matching accuracy, **85%** time saved in pre-screening, **78%** interview success rate.

Mathematics Instructor (Freelance) Jan – Jun 2025

Self-employed · Tétouan, Morocco

- Tutored high-school students in algebra, functions, analysis and geometry, adapting lessons to each student's level and preparing them for exams.

EDUCATION

Engineering Degree — Big Data & Artificial Intelligence 2024 – Present

National School of Applied Sciences of Tétouan (ENSA Tétouan)

Integrated Preparatory Program for Engineering Schools 2022 – 2024

ENSA Tétouan

Scientific Baccalaureate — French Option 2020 – 2022

Tarik Ibn Ziad High School, Kelaat M'Gouna

SELECTED PROJECTS

Case studies, code and demos: **abdessamadaitbrahim.github.io/projects.html**

- **Vision Vocab Tutor** (YOLOv8, vision LLM, Flask): web app that teaches English vocabulary from real images: YOLOv8 detects objects, a vision LLM writes quiz questions, difficulty adapts to the learner and a spaced-repetition (SM-2) bank schedules review.
- **Facial Emotion Recognition** (PyTorch, CNN, VGG, ResNet): VGG and ResNet built from scratch on 49,779 images under the same protocol; ResNet reached **75.4%** test accuracy vs 66.7% for VGG with **34× fewer parameters**.
- **Image Captioning** (TensorFlow, CNN, LSTM, Transformer): CNN encoders with LSTM, BiLSTM and Transformer decoders on Flickr30K, evaluated with BLEU, compared with a BLIP model and served through a Flask app.
- **Heart Guardian** (ESP32, XGBoost, Unity): real-time ECG pipeline (Pan-Tompkins R-peak detection, HRV features, XGBoost classifier) that adapts a game to the player's cardiac state.
- **SportLineBot** (ESP32-S3, Flask, NLP): autonomous line-marking robot controlled by French natural-language commands (TF-IDF + Naive Bayes, ∼90% accuracy) with Isolation Forest anomaly detection.
- **SSH Log Monitoring with AI** (Isolation Forest, Flask): real-time anomaly detection on SSH logs with automatic IP blocking, email alerts and a dashboard.
- **BI Data Warehouse** (SSIS, SSAS, Power BI): star-schema warehouse with SSIS ETL, SSAS OLAP cube on SQL Server and Power BI dashboards.

HACKATHONS & COMPETITIONS

- **Water4Future International Hackathon 2026**, Montpellier (UNESCO ICIREWARD): **semi-finalist, top 38 of 125+ teams** and **Water Talent Prize** for AMAN, an AgriTech solution with a hydroponic IoT system, a real-time dashboard and an AI diagnostic chatbot.
- **Game4Health Hackathon 2026**, Rabat (SmartTech Lab, iSMAGI): **finalist** with Heart Guardian, pitched with a business model and a 6-month roadmap.
- **MECA WORLD CUP 2026**, national robotics competition: developed the AI module of SportLineBot, an autonomous field-marking robot.

TECHNICAL SKILLS

Programming Languages:	Python, SQL, C++
ML & Deep Learning:	PyTorch, TensorFlow, scikit-learn, XGBoost, Isolation Forest, CNN (VGG, ResNet), LSTM/GRU, Transformers, model evaluation
NLP & LLMs:	Hugging Face Transformers, fine-tuning (XLM-RoBERTa, mBERT, DarijaBERT), multilingual NLP, LLM agents (Dify), prompt engineering
Computer Vision:	OpenCV, YOLOv8, image captioning, facial emotion recognition
Data Engineering & Big Data:	pandas, NumPy, PySpark, Apache Kafka, Hadoop MapReduce, ETL/ELT, data cleaning and validation, feature engineering
Backend & APIs:	FastAPI, Flask, REST APIs
Databases, Cloud & Tools:	PostgreSQL, MySQL, MariaDB, SQL Server, SQLite, MongoDB, AWS (EC2, S3), Docker, Git, Linux, ESP32
Business Intelligence:	Power BI, Tableau, SSIS, SSAS, Streamlit

LANGUAGES

Tamazight (Native) • **Arabic** (Professional) • **French** (Professional) • **English** (Professional) • **Spanish** (Elementary)